

- Every entry has the function name in the middle of it with the same index number; i.e., *main* with value 1 in *index 1* field. So you can easily see the cycle for it, in *Index 1* field kernel operations calling *main*, then *main* calling *func3* and so on. The same follows for the rest of the index.
- In *Index 3*, *func1* is called by *main* with 399 calls; spent 0.02 time in children process (this is substantiated by 0.02 in *func2* call). *func3* is directly executed from *main*.
- The middle line in each section is referenced by a unique index number. Above the line indicates originating function for the function given in index number and below the line indicates the process call forwarded to the next function. For example, in index 3 section, the call originated from *main()* function to *func1()* and then to *func2()*. Here *func1()* has index number 3.
- *<spontaneous>* indicates that the parent function cannot be determined, generally due to system-related APIs.
- A cumulative time recording implies that 98 per cent of the total time was spent in performing *func3* operations, as per *index 2*.

Function calls substantiate that the *filegen()* function is called thrice for a single call of *func2()*. *func1()* and *func2()* are called for the same time; the counter values present in the code prove this. Counters are displayed after successful execution of the code. *func1()* and *func2()* functions comprise the same timings for calculations. *func3* is called for 9601 times. The call-graph gives the call-tree data.

Source code annotations in profiling

To generate annotations in the source code presented, the source code has to be included in the executable while compiling code, using the 'g' switch:

```
gcc gprof_test_code_svp.c -g -pg -lc
```

Now run the code normally, and use the source code highlighting option '-A' in Gprof to have annotations marked in the source code. This gives line-by-line profiling, which is easy to understand.

```
gprof a.out gmon.out -A
*** File /root/gprof_test_code_svp.c:
/*
    Name :Subodh Pachghare
    Author :I am the author!!!
    Description :Function calling test program
for gprof
*/
#include <stdio.h>
```

```
int counter1=0;          /*Global Variable*/
int counter2=0;          /*Global Variable*/
int counter3=0;          /*Global Variable*/
int main()
#### -> {
    int p=0;
    system("> /root/hashes.txt"); /*File will be
overwritten or Created*/
    system("> /root/tempc.txt"); /*File will be
overwritten or Created*/
    printf("Initial value in Main function=
%d\n",p);
    while(p<10000) /*Condition for while loop*/
    {
        if(p>9600 && p<10000)
        {
            func1();
        }
        else
        {
            func3();
        }
        p++;
    }
    printf("Last value in Main function (value(p
incremented for) = %d\n",p);
    printf("Func1(),Func2(),Func3() & filegen()
called for %d %d %d times\n",counter1,counter1,counter3,coun
ter2);
}
func1()
399 -> {
    int j=0;
    while(j<66667) /*Another operation to
eat CPU cycles*/
    {
        j++;
        if(j==66668)
        {
            counter1++;
        }
    }
    func2(); /*Call to Function func2()*/
}
func2()
399 -> {
    int m=0;
    for (m=1;m<1000000;m++)
    {
        if(m<4)
        {
            filegen(); /*Call to Function
filegen()*/
            counter2++;
        }
    }
}
```